

COSC 594

DM for Offline RL

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Announcement

Check Canvas for Project Guideline
Class presentation from 11/3 to 11/19

Class participation

#1 Scribing lectures

#2 Providing feedback

Outline

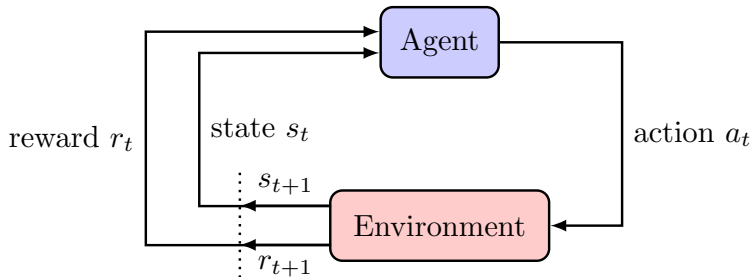
- RECAP
- OFFLINE POLICY OPTIMIZATION
- DIFFUSION POLICY OPTIMIZATION
- POLICY OPTIMIZATION PERSPECTIVE IN DISTRIBUTION SPACE

Outline

■ RECAP

RL Framework

■ MARKOV DECISION PROCESSES (MDPs)



Goal $\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s_0} [V^{\pi}(s_0)]$	$\Leftrightarrow$	Direct policy search $\pi^+ \leftarrow \pi + \nabla_{\pi} V^{\pi}$
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π – a policy that maps states to actions

$V^{\pi}(s_0) := \mathbb{E} [\text{sum of all rewards received}]$

Policy optimization

Policy: $\pi : S \text{ (states)} \rightarrow \Delta_A \text{ (action distribution)}$

Value function: $V^\pi(\rho) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \rho \right]$

$P(\cdot \mid s, a) : S \times A \rightarrow \Delta_A$ – transition kernel

$s_0, a_0, r_0, s_1, a_1, \dots$ – a trajectory τ generated by a policy π

■ NONCONVEX OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad V^\pi(\rho)$$

★ Search the decision in [a function space](#)

Policy parametrization

Policy: $\pi_{\theta}(\cdot | s): S \rightarrow \Delta_A$

Parameter: $\theta \in \mathbb{R}^d$

■ LINEAR SOFTMAX POLICY

$$\pi_{\theta}(a | s) = \frac{e^{\theta^{\top} \phi_{s,a}}}{\sum_{a'} e^{\theta^{\top} \phi_{s,a'}}} \text{ w/ feature } \phi_{s,a} \in \mathbb{R}^d$$

■ GENERAL SOFTMAX POLICY

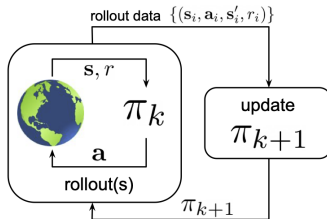
$$\pi_{\theta}(a | s) = \frac{e^{f_{\theta}(s,a)}}{\sum_{a'} e^{f_{\theta}(s,a')}} \text{ w/ network } f_{\theta}(s,a)$$

■ GAUSSIAN POLICY

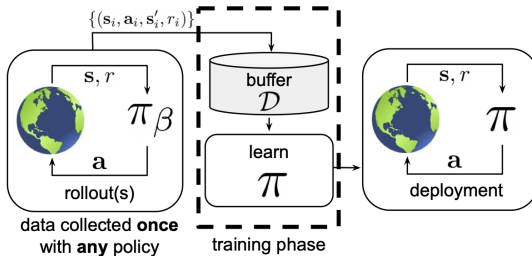
$$\pi_{\theta}(a | s) = \mathcal{N}(a; \mu_{\theta}(s), \Sigma(s)) \text{ w/ mean } \mu_{\theta}(s), \text{ covariance } \Sigma(s)$$

Online vs offline RL

■ ONLINE RL



■ OFFLINE RL



Outline

- RECAP
- OFFLINE POLICY OPTIMIZATION

Off-policy policy gradient

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^{\pi_\theta}(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi_\theta(\cdot | s)} [Q^{\pi_\theta}(s, a)]$

■ ONLINE POLICY GRADIENT

$$\nabla_\theta V^{\pi_\theta}(\rho) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_\theta}(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t) \right]$$

★ expensive on-policy samples

■ OFF-POLICY POLICY GRADIENT

$$\nabla_\theta V^{\pi_\theta}(\rho) = \mathbb{E}_{\tau \sim \pi_\beta} \left[\frac{\pi_\theta(\tau)}{\pi_\beta(\tau)} \sum_{t=0}^{\infty} \gamma^t Q^{\pi_\theta}(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t) \right]$$

Unbiased w/ **high variance**

Actor-critic: policy evaluation

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^{\pi_\theta}(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi_\theta(\cdot | s)} [Q^{\pi_\theta}(s, a)]$

■ POLICY EVALUATION

$$Q^{\pi_\theta} \approx Q^{\phi^+} \quad \text{w/ parameter } \phi^+$$

$$\phi^+ \in \underset{\phi}{\operatorname{argmin}} \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \left[\left(r(s, a) + \gamma Q^{\bar{\phi}}(s', a') - \gamma Q^{\phi}(s, a) \right)^2 \right]$$

$Q^{\bar{\phi}}$ – target network w/ parameter $\bar{\phi}$

$a' \sim \pi_\theta(\cdot | s')$ – next action

Minimization of temporal difference error

Actor-critic: policy improvement

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^{\pi_\theta}(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi_\theta(\cdot | s)} [Q^{\pi_\theta}(s, a)]$

■ POLICY IMPROVEMENT

$$\begin{aligned} & \underset{\theta}{\text{maximize}} \quad V^{\pi_\theta}(\rho) \\ & \quad \Downarrow \\ & \underset{\theta}{\text{maximize}} \quad \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot | s)} [Q^\phi(s, a)] \end{aligned}$$

■ OFF-POLICY POLICY GRADIENT

$$\mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot | s)} [Q^\phi(s, a) \nabla \log \pi_\theta(\cdot | s)]$$

Biased w/o importance weight

Actor-critic learning

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^{\pi_\theta}(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi_\theta(\cdot | s)} [Q^{\pi_\theta}(s, a)]$

■ POLICY EVALUATION

$$\phi^+ \in \underset{\phi}{\operatorname{argmin}} \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \left[\left(r(s, a) + \gamma Q^{\bar{\phi}}(s', a') - \gamma Q^\phi(s, a) \right)^2 \right]$$

$Q^{\bar{\phi}}$ – target network w/ parameter $\bar{\phi}$

$a' \sim \pi_\theta(\cdot | s')$ – next action

■ POLICY IMPROVEMENT

$$\theta^+ \in \underset{\theta}{\operatorname{argmax}} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot | s)} [Q^{\phi^+}(s, a)]$$

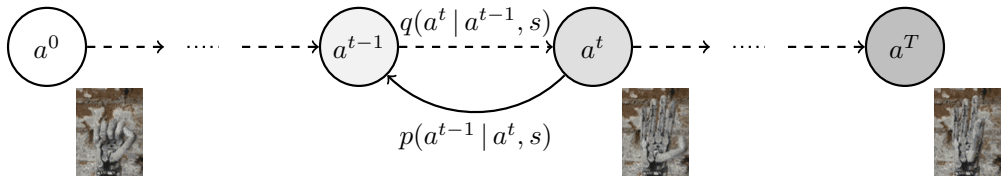
$$\theta \leftarrow \eta \theta + (1 - \eta) \theta^+, \phi \leftarrow \eta \phi + (1 - \eta) \phi^+$$

Out-of-distribution actions $\implies$ Overestimated value function

Outline

- RECAP
- OFFLINE POLICY OPTIMIZATION
- DIFFUSION POLICY OPTIMIZATION

Diffusion policy



data point a^0 – action

■ ANALOGUE W/ VAE

★ noiser (encoder)

$$q(a^{1:T} | a^0, s) = q(a^1 | a^0, s) q(a^2 | a^1, s) \dots q(a^T | a^{T-1}, s)$$

★ denoiser (decoder)

$$p(a^0 | a^{1:T}, s) = p(a^T, s) p(a^{T-1} | a^T, s) \dots p(a^0 | a^1, s) \implies \pi(a^{0:T} | s) \text{ is a RL policy!}$$

■ GROUND-TRUTH DENOISER

$$q(a^{t-1} | a^t, a^0, s) = \mathcal{N}(a^{t-1}; \mu(a^t, a^0, s), \sigma^2(t)I)$$

$$q(a^t | a^{t-1}, s) = \mathcal{N}(a^t; \sqrt{\alpha_t}a^{t-1}, (1 - \alpha_t)I)$$

$$\mu(a^t, a^0, s) = \frac{1}{\sqrt{\alpha_t}}a^t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_0 - \text{noise-based mean}$$

$\sigma^2(t)$ – variance

■ APPROXIMATE DENOISER

$$\pi_\theta(a^{t-1} | a^t, s) = \mathcal{N}(a^{t-1}; \mu_\theta(a^t, t, s), \sigma^2(t)I), \quad \mu_\theta(a^t, t, s) = \frac{1}{\sqrt{\alpha_t}}a^t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_\theta(a^t, t, s)$$

■ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(q(a^{t-1} | a^t, a^0, s) \parallel \pi_\theta(a^{t-1} | a^t, s)) \quad \Leftrightarrow \quad \underset{\theta}{\text{minimize}} \quad \|\epsilon_0 - \epsilon_\theta(a^t, t, s)\|^2$$

Cloned policy $\pi_{\theta^\star}(a^{t-1} | a^t, s)$

Diffusion policy optimization

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^{\pi_\theta}(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi_\theta(\cdot | s)} [Q^{\pi_\theta}(s, a)]$

■ POLICY IMPROVEMENT

$$\theta^+ \in \operatorname{argmax}_{\theta} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot | s)} [Q^{\phi^+}(s, a)]$$

Out-of-distribution actions

$\Downarrow$

$$\theta^+ \in \operatorname{argmax}_{\theta} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot | s)} [Q^{\phi^+}(s, a)] - \underbrace{\frac{1}{\eta} \mathbb{E}_{t, \epsilon_0, (s, a^0) \sim \mathcal{D}} [\|\epsilon_0 - \epsilon_\theta(a^t, t, s)\|^2]}_{\text{Behavior cloning}}$$

■ REPARAMETRIZATION TRICK

$$q(a^t | a^{t-1}, s) = \mathcal{N}(a^t; \sqrt{\alpha_t} a^{t-1}, (1 - \alpha_t)I)$$

$\Downarrow$

$$q(a^t | a^0, s) = \mathcal{N}\left(a^0; \sqrt{\bar{\beta}_t} a^0, (1 - \bar{\beta}_t)I\right)$$

$$\beta_t = 1 - \alpha_t, \bar{\beta}_t = \prod_{s=1}^t \beta_s$$

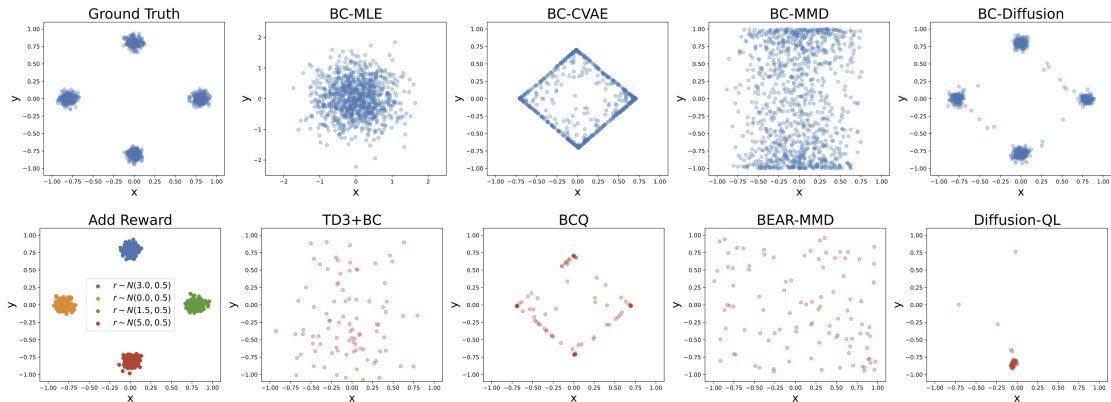
$\Updownarrow$

$$\boxed{a^t = \sqrt{\bar{\beta}_t} a^0 + \sqrt{1 - \bar{\beta}_t} \epsilon \quad \text{for } \epsilon \sim \mathcal{N}(0, I)}$$

★ approximate action

$$a \sim \pi_\theta(\cdot | s) \approx a^0 = \frac{1}{\sqrt{\bar{\beta}_t}} a^t - \frac{\sqrt{1 - \bar{\beta}_t}}{\sqrt{\bar{\beta}_t}} \epsilon_\theta(a^t, t, s)$$

A simple bandit task



Diffusion policy limits **in-distribution actions**

Expressiveness of diffusion policy as **a multi-modal distribution**

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- DIFFUSION POLICY OPTIMIZATION
- POLICY OPTIMIZATION PERSPECTIVE IN DISTRIBUTION SPACE

Policy optimization perspective in distribution space

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^\pi(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi(\cdot | s)} [Q^\pi(s, a)]$

■ PERFORMANCE DIFFERENCE

$$\begin{aligned} V^\pi(\rho) - V^{\pi_\beta}(\rho) &= \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\rho^\pi, a \sim \pi(\cdot | s)} [A^{\pi_\beta}(s, a)] \\ &\approx \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\rho^{\pi_\beta}, a \sim \pi(\cdot | s)} [A^{\pi_\beta}(s, a)] \end{aligned}$$

$\Downarrow$

$$\underset{\pi}{\text{maximize}} \quad V^\pi(\rho) \quad \approx \quad \underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s \sim d_\rho^{\pi_\beta}, a \sim \pi(\cdot | s)} [Q^{\pi_\beta}(s, a)]$$

Distribution shift from d_ρ^π to $d_\rho^{\pi_\beta}$

Optimal policy

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^\pi(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi(\cdot | s)} [Q^\pi(s, a)]$

■ KL-REGULARIZED POLICY IMPROVEMENT

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s \sim d_\rho^{\pi_\beta}, a \sim \pi(\cdot | s)} [Q^{\pi_\beta}(s, a)] - \underbrace{\frac{1}{\eta} \mathbb{E}_{s \sim d_\rho^{\pi_\beta}} [D_{\text{KL}}(\pi(\cdot | s) \parallel \pi_\beta(\cdot | s))]}_{\text{Behavior cloning}}$$

★ optimal solution

$$\pi^*(\cdot | s) \propto \pi_\beta(\cdot | s) e^{\eta Q^{\pi_\beta}(s, \cdot)} \quad \text{for } s \in \text{supp}(d_\rho^{\pi_\beta})$$

Recovery of diffusion policy optimization

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^\pi(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi(\cdot | s)} [Q^\pi(s, a)]$

■ KL-REGULARIZED POLICY IMPROVEMENT

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s \sim \mathbf{d}_\rho^{\pi_\beta}, a \sim \pi(\cdot | s)} [Q^{\pi_\beta}(s, a)] - \underbrace{\eta \mathbb{E}_{s \sim \mathbf{d}_\rho^{\pi_\beta}} [D_{\text{KL}}(\pi(\cdot | s) \parallel \pi_\beta(\cdot | s))]}_{\text{Behavior cloning}}$$

$\Downarrow \quad \pi \leftarrow \pi_\theta, \mathbf{d}_\rho^{\pi_\beta} \leftarrow \mathcal{D}, Q^{\pi_\beta} \leftarrow Q^{\phi^+}, \text{ point-wise KL} \leftarrow \text{path-wise KL}$

$$\theta^+ \in \underset{\theta}{\text{argmax}} \mathbb{E}_{s \sim \mathbf{D}, a \sim \pi_\theta(\cdot | s)} [Q^{\phi^+}(s, a)] - \underbrace{\eta \mathbb{E}_{t, \epsilon_0, (s, a^0) \sim \mathbf{D}} [\|\epsilon_0 - \epsilon_\theta(a^t, t, s)\|^2]}_{\text{Behavior cloning}}$$

Policy optimization w/ experience replay

Offline data: $\mathcal{D}^k = \{(s_t^k, a_t^k, s_{t+1}^k)\}$ for k th behavior policy π_β^k

Value function: $V^\pi(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi(\cdot | s)} [Q^\pi(s, a)]$

■ PERFORMANCE DIFFERENCE

$$\begin{aligned} V^\pi(\rho) - \sum_{k=1}^K w_k V^{\pi_\beta^k}(\rho) &= \frac{1}{1-\gamma} \sum_{k=1}^K w_k \mathbb{E}_{s \sim d_\rho^\pi, a \sim \pi(\cdot | s)} [A^{\pi_\beta^k}(s, a)] \\ &\approx \frac{1}{1-\gamma} \sum_{k=1}^K w_k \mathbb{E}_{s \sim d_\rho^{\pi_\beta^k}, a \sim \pi(\cdot | s)} [A^{\pi_\beta^k}(s, a)] \end{aligned}$$

$\Downarrow$

$$\underset{\pi}{\text{maximize}} \quad V^\pi(\rho) \quad \approx \quad \underset{\pi}{\text{maximize}} \quad \sum_{k=1}^K w_k \mathbb{E}_{s \sim d_\rho^{\pi_\beta^k}, a \sim \pi(\cdot | s)} [Q^{\pi_\beta^k}(s, a)]$$

Distribution shift from d_ρ^π to $\sum_{k=1}^K w_k d_\rho^{\pi_\beta^k}$

Optimal policy

Offline data: $\mathcal{D}^k = \{(s_t^k, a_t^k, s_{t+1}^k)\}$ for k th behavior policy π_β^k

Value function: $V^\pi(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi(\cdot | s)} [Q^\pi(s, a)]$

■ KL-REGULARIZED POLICY IMPROVEMENT

$$\underset{\pi}{\text{maximize}} \quad \sum_{k=1}^K w_k \mathbb{E}_{s \sim d_\rho^{\pi_\beta^k}, a \sim \pi(\cdot | s)} [Q^{\pi_\beta^k}(s, a)] - \underbrace{\frac{1}{\eta} \sum_{k=1}^K \mathbb{E}_{s \sim d_\rho^{\pi_\beta^k}} [D_{\text{KL}}(\pi(\cdot | s) \| \pi_\beta^k(\cdot | s))]}_{\text{Behavior cloning}}$$

★ optimal solution

$$\pi^*(\cdot | s) \propto \prod_{k=1}^K \pi_\beta^k(\cdot | s) e^{(\eta/K) \sum_{k=1}^K w_k Q^{\pi_\beta^k}(s, \cdot)} \quad \text{for } s \in \text{supp}(d_\rho^{\pi_\beta^1}, \dots, d_\rho^{\pi_\beta^K})$$

Diffusion policy optimization

Offline data: $\mathcal{D}^k = \{(s_t^k, a_t^k, s_{t+1}^k)\}$ for k th behavior policy π_β^k

Value function: $V^\pi(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi(\cdot | s)} [Q^\pi(s, a)]$

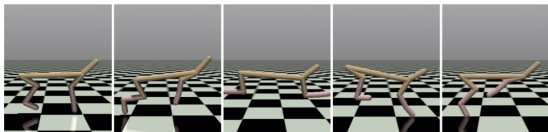
■ KL-REGULARIZED POLICY IMPROVEMENT

$$\underset{\pi}{\text{maximize}} \quad \sum_{k=1}^K w_k \mathbb{E}_{s \sim d_\rho^{\pi_\beta^k}, a \sim \pi(\cdot | s)} [Q^{\pi_\beta^k}(s, a)] - \underbrace{\frac{1}{\eta} \sum_{k=1}^K \mathbb{E}_{s \sim d_\rho^{\pi_\beta^k}} [D_{\text{KL}}(\pi(\cdot | s) \| \pi_\beta^k(\cdot | s))]}_{\text{Behavior cloning}}$$

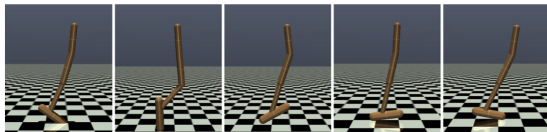
$\Downarrow \quad \pi \leftarrow \pi_\theta, d_\rho^{\pi_\beta^k} \leftarrow \mathcal{D}^k, Q^{\pi_\beta^k} \leftarrow Q^{\phi_k^+}, \text{ point-wise KL} \leftarrow \text{path-wise KL}$

$$\theta^+ \in \underset{\theta}{\text{argmax}} \quad \sum_{k=1}^K w_k \mathbb{E}_{s \sim \mathcal{D}^k, a \sim \pi_\theta(\cdot | s)} [Q^{\phi_k^+}(s, a)] - \underbrace{\frac{1}{\eta} \sum_{k=1}^K \mathbb{E}_{t, \epsilon_0^k, (s, a^0) \sim \mathcal{D}^k} [\|\epsilon_0^k - \epsilon_\theta^k(a^t, t, s)\|^2]}_{\text{Behavior cloning}}$$

Four OpenAI Gym tasks



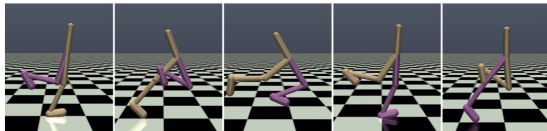
(a) HalfCheetah-v2



(b) Hopper-v2



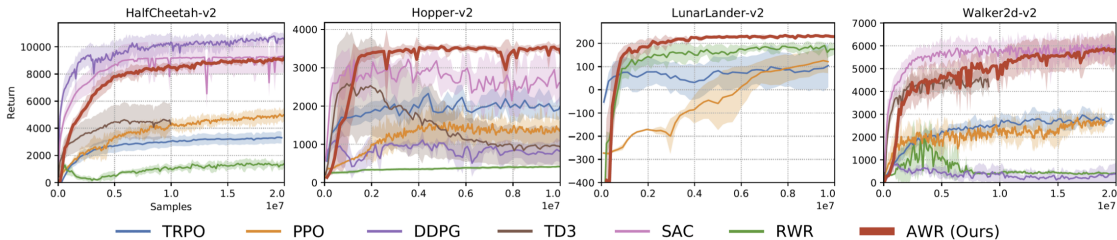
(c) LunarLander-v2



(d) Walker2d-v2

example source: [arXiv:1910.00177](https://arxiv.org/abs/1910.00177)

Four OpenAI Gym tasks



Exploitation via **experience replay**

Summary

■ RECAP

Policy optimization, offline vs online RL

■ OFFLINE POLICY OPTIMIZATION

Off-policy policy gradient, actor-critic learning

■ DIFFUSION POLICY OPTIMIZATION

Diffusion policy, diffusion policy optimization, policy optimization in distribution space

■ POLICY OPTIMIZATION PERSPECTIVE IN DISTRIBUTION SPACE

Policy optimization in distribution space, experience replay